

JOANNE HOAR

Software Developer & Technology Educator

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Hanna, Alberta, Canada • Remote

Summary

Canadian Software Developer and Technology Educator with a Master of Science in Computer Science (University of Calgary, including graduate-level HCI coursework) and over twenty years of continuous experience building user-facing systems. HCI has been the common thread across every role: translating CLIs into desktop GUIs at the start of a career, designing streaming media interfaces and interactive web applications mid-career, building cross-platform widget libraries and demo apps across seven UI frameworks at an SDK company, and now teaching front-end UI development to the next generation of developers online. Committed to accessible, student-centered teaching and to connecting HCI theory to real professional practice.

Core Competencies

- HCI Principles & Interaction Design
- Front-End UI Development (Angular, React)
- Game Interface Design & Player Experience
- Augmented Reality (ARKit) & 3D Graphics
- Accessibility & Inclusive Design
- Online Course Delivery (Brightspace / D2L)
- Technical Curriculum Design & Assessment
- Live Coding, Mentorship & Student Feedback

Teaching Experience

Technology Instructor

Jan 2025 – Present

Robertson College (contract) — Online Instructor-Guided (Brightspace / D2L)

- Teach 12 subjects for **Full Stack Development** and **Data Analytics** diploma programs, with 20 course offerings taught since January 2025, including *Front End Frameworks* (Angular & React), *Web Development Basics*, *ASP.NET Core*, *JavaScript*, *C# Programming*, *Object-oriented Programming*, *Version Control*, and more.
- Conduct 3+ synchronous live coding sessions per week; model real-time debugging and problem-solving to demystify professional practice.
- Design supplemental code examples, reference sheets, and video walkthroughs to support diverse learning styles; publish materials as open-source on GitHub.
- Earned average student satisfaction rating of **4.5/5.0**; feedback consistently highlights approachability, real-world teaching style, and responsiveness.

Angular, React, ASP.NET Core, JavaScript, C#, Java, Git, GitHub, Azure, SQL, Brightspace

Research / Teaching Assistant

2001 – 2004

VLab Computer Graphics Group / AI Research Hub, University of Calgary

- Served as TA for first-year introductory and advanced third-year computer science courses.
- Supplemented instructor lesson plans with independent presentations and code/math walk-throughs; received positive student feedback.
- Participated in interdepartmental research projects and contributed to five published papers.

C++, OpenGL, PHP, Linux, Solaris

HCI-Relevant Industry Experience

Senior Software Developer

2022 – 2025

bitHeads Inc. — brainCloud BaaS

- Designed and built demonstration applications and **widget libraries for every supported client platform** to showcase SDK capabilities — a cross-platform UI portfolio spanning **Dear ImGui** (C++), **Unreal Engine UMG**, **SwiftUI/UIKit** (iOS), **Android Views** (Java/Kotlin), **Unity** (C#), and **PlayStation SDK**.
- Maintained developer-facing APIs for 50+ published games across 12 platforms; API design requires the same HCI principles as any interface — affordance, discoverability, error recovery, and minimal surprise.
- Authored onboarding documentation and sample projects engineered to minimize cognitive load for incoming client developers.
- Hired in part for educational background — role required teaching clients one-on-one, strong documentation skills, and developing accessible examples and demonstrations.
- Supported integration for notable titles including *SUPERHOT VR*, *Tony Hawk's Skate Jam*, and *Crypt of the NecroDancer* across 12 platforms.

Unreal Engine, C++, iOS, Android, PlayStation, Nintendo Switch, Xbox, Node.js, C# (Unity, Godot)

Co-founder / Lead Developer / Instructor

Silicon Hanna Inc. (self-employed; founded as SuRJE Software Solutions Inc.; concurrent with BJ Pipeline 2005–08 and Fekete Associates Inc. 2008–2011)

- Designed and shipped eleven mobile games; interaction design, touch interface usability, and player-experience feedback loops were central to every title.
- Developed augmented reality applications using **ARKit**, designing novel spatial interfaces for consumer mobile devices.
- Instructed intermediate C++ programming for the **University of Calgary Department of Continuing Education**; developed curriculum, lectures, sample code, and assignments.

Unreal Engine, Objective-C, Swift, iOS, Android, ARKit, Augmented Reality, C++

Tools Programmer

2008 – 2011

Fekete Associates Inc. (S&P Global)

- Contributed to the company's shared **MFC UI tools library** — the consistent component foundation used across all internal development projects; brought UI coherence and interaction consistency to the full product suite.

- Built 3D data visualization tools (OpenGL) for pipeline analysis products; clarity of spatial information presentation was a direct safety concern.
- Designed and delivered a voluntary OpenGL graphics elective for colleagues; students built an interactive OpenGL widget within the MFC framework as a capstone project; adapted for both beginners and senior developers.

C++, OpenGL, MFC, Windows

Intermediate Software Engineer

2005 – 2008

BJ Pipeline Inspection Services Inc. (Baker Hughes)

- Designed 3D OpenGL visualizations (**Vectra, Gemini**) for professional pipeline analysts; spatial data clarity was a safety-critical design requirement.
- Ported an Excel-based inspection reporting workflow to a full **LAMP** web application with interactive JavaScript forms, dynamic charts, and interactive geographic maps — replacing a static spreadsheet with a live, accessible tool for non-technical stakeholders.

C++, OpenGL, PHP, JavaScript, SQL, LAMP

Junior Software Engineer

2004 – 2005

Codefast.com

- Primary mandate: build **graphical user interfaces for existing CLI tools** — the definitive HCI task of making complex command-line workflows accessible to broader users.
- Ported applications across GUI frameworks (**Qt, Wx**); developed UIs for software lifecycle automation tooling.

C++, Qt, Wx, Shell Script, Windows, Unix

Software Researcher / Lead Developer

2005 – 2006

PlayStarMusic Corp. (concurrent with BJ Pipeline)

- Designed the end-user interface for a media streaming storefront: browsing, playback controls, purchase flow, and seamless DRM integration that kept the listening experience uninterrupted.
- Lead developer and system administrator for the full platform stack.

C++, PHP, JavaScript, .NET, Windows, Linux

Education

- **M.Sc. Computer Science** – University of Calgary 2004
Thesis: Agent-based Development of Natural Transportation Networks
Graduate coursework included Human-Computer Interaction, Computer Graphics, and Artificial Intelligence
- **B.Sc. Computer Science** – University of Calgary 2001

Selected Publications

- Hoar, J., et al. "Bacterial Chemotaxis in Silico," *Australian Conference on Artificial Life (ACAL)*, 2003 *Best Student Paper Award*
- Hoar, R., Jacob, C. "The application of artificial intelligence to transportation system design," *XRDS: Crossroads, ACM Magazine for Students*, 2003

Honors & Awards

- Departmental Nomination: Governor General's Gold Medal, Faculty of Graduate Studies 2005
- Postgraduate Scholarship NSERC D3, Natural Sciences and Engineering Research Council of Canada 2005
- Graduate Student Scholarship, Alberta Learning 2004
- Best Student Paper Award (Co-author), ACAL 2003
- CPSC Departmental Research Award, University of Calgary 2003

Technical Skills

HCI & UI: Interaction Design, User Experience, Accessible Design, Touch Interfaces, Augmented Reality, 3D Visualization, Front-End Frameworks

Languages: C++, C#, JavaScript, TypeScript, Java, Objective-C, Swift, Kotlin, PHP, SQL

Frameworks & Tools: Angular, React, ASP.NET Core, Unreal Engine, OpenGL, ARKit, Node.js, Azure, Git

Teaching Tools: Brightspace / D2L, GitHub (open-source course materials), Synchronous & Asynchronous Online Delivery

Platforms: iOS, Android, macOS, Windows, Linux, PlayStation, Nintendo Switch, Xbox

Languages (Human): English (native), French (intermediate), Spanish (rudimentary)

Published Interactive Applications

Silicon Hanna Inc. (founded as SuRJE Software Solutions Inc.) — 11 titles, 2008–2022: Ragin' Bull, AR Picture Portal, Fragment Reality, IceCream AR, Jewel Slide, Word Slide & Find, A Word Theme, CryptoQuip, Rafter's Loot, Star Chase, Bitsy

Community Outreach & Leadership

- Computer Programming for Kids, J.C. Charyk School 2019–2021
- SCIBerMENTOR Correspondent / Mentor 2002–2005
- President, Computer Science Graduate Society 2002–2003
- VP Academic, Computer Science Undergraduate Society 2001
- Treasurer, Hanna & District Minor Soccer Association 2025–Present
- Vice President, Hanna Figure Skating Club 2024–Present

Work Samples

 github.com/xcodegirl/career – [xcodegirl-work-samples.md](#)